

Nakshatra Jain

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Summary

Final-year CS Engineer at VIT Bhopal (CGPA: **8.65/10**), focused on backend systems and cloud infrastructure. Comfortable building services with Spring Boot, Kafka, and Redis, and deploying them on AWS with GitHub Actions. Recent projects cover multi-tenant SaaS, real-time analytics, and semantic search. OCI Architect Associate & DevOps Professional certified.

Technical Skills

Languages: Java, Python, C++, SQL, JavaScript

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, Microservices, JWT, RBAC, WebSockets, Swagger/OpenAPI

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, IAM, CloudWatch), Docker, Terraform, GitHub Actions, CI/CD, Render, Vercel

Data & Messaging: PostgreSQL, MySQL, MongoDB, Redis, Apache Kafka, Query Optimization, Indexing

Frontend: React.js, Next.js, Tailwind CSS, Firebase Authentication

Observability & Testing: Prometheus, Grafana, Micrometer, JUnit 5, Mockito, Postman, k6

AI/Search: LangChain, ChromaDB, Qdrant, OpenAI APIs, RAG Pipelines, Vector Search

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Projects

NexGate — Multi-Tenant API Gateway with Real-Time Traffic Analytics

Apr 2026 – May 2026

Java, Spring Boot, Apache Kafka, Redis, PostgreSQL, WebSockets, Prometheus, Grafana, Docker, Next.js, Render, Vercel, GitHub Actions

Links: GitHub | Live

- Built a multi-tenant API gateway in Spring Boot handling request proxying, API key auth, and per-tenant rate limiting via a **Redis sliding-window algorithm using atomic Lua scripts**. Load-tested locally with k6 (90 VUs): **980 req/min, 62ms** median latency, 0% errors across 3 simulated tenants.
- Moved request/response logging off the hot path into an async **Kafka** pipeline writing to PostgreSQL, so gateway latency stays unaffected by logging volume; added dead-letter topic handling with 3-retry backoff for failed writes.
- Built a live traffic dashboard over **WebSockets** showing per-tenant request rate, latency, and error counts, backed by a **Prometheus + Grafana** stack with custom Micrometer metrics for gateway throughput and p99 latency.

VaultAI — Semantic Code & Document Search Platform

Feb 2026 – Mar 2026

Spring Boot, FastAPI, Next.js, Qdrant, Celery, Redis, PostgreSQL, Docker Compose, OpenAI

Links: GitHub | Live

- Split the system into a **Spring Boot** gateway and a **FastAPI** embedding service; ingested repos via GitHub OAuth and chunked code with **tree-sitter** at function/class boundaries instead of naive line splitting.
- Combined **BM25** keyword search with **Qdrant** vector search and a cross-encoder re-ranking step, so results aren't purely keyword- or embedding-driven; cached repeated queries in Redis to cut redundant embedding calls.
- Secured the API with JWT auth and scoped API keys, added Redis-based rate limiting, and wired up Prometheus/Grafana for query latency and cache-hit visibility.

Nexora — Multi-Tenant Project Management SaaS

Dec 2025 – Jan 2026

Spring Boot, React, NeonDB (PostgreSQL), Firebase Auth, Docker, Render, Vercel, GitHub Actions

Links: GitHub | Live

- Wrote tenant-scoped JPA queries to enforce data isolation between organizations; confirmed **zero cross-tenant leakage** across 5 test organizations with JUnit 5 integration tests.
- Traced a slow endpoint to Hibernate **N+1** queries from **@OneToMany** serialization; fixed it with DTO projections and fetch-join optimization, cutting DB round-trips **70%** and local benchmark response time from **450ms to 130ms**.
- Secured 15+ REST APIs with Firebase Auth, Spring Security JWT, and 3-tier RBAC (OWNER/ADMIN/MEMBER); documented with Swagger/OpenAPI and shipped through a GitHub Actions CI/CD pipeline.

Education

VIT Bhopal University

Aug 2023 – May 2027

B.Tech in Computer Science Engineering — Cloud Computing & Automation **CGPA: 8.65/10**

Bhopal, MP

Certifications

Oracle Cloud: OCI Architect Associate | OCI DevOps Professional | OCI Foundations Associate

IBM: DevOps Fundamentals

AWS (In Progress): Solutions Architect Associate (SAA-C03) | Cloud Practitioner (CLF-C02)